

6. Despite safety precautions, industrial accidents occur.

- (a) Methyl isocyanate (MIC) was involved in the Bhopal disaster that led to the deaths of thousands of people. A highly toxic cloud of MIC was released from chimneys at the chemical plant. On the night of the disaster it reacted with water that leaked into a storage tank, causing a valve to blow.

The structural formula of MIC is  $\text{H}_3\text{C}-\text{N}=\text{C}=\text{O}$ .

- (i) Complete the table below to calculate the total energy required to break all the bonds in 1 mole of the reactants, MIC and water. [5]

| $\text{H}_3\text{C}-\text{N}=\text{C}=\text{O}$ |                 |                                   |                                                |
|-------------------------------------------------|-----------------|-----------------------------------|------------------------------------------------|
| Bond type                                       | Number of bonds | Energy to break one bond (kJ/mol) | Energy to break all of each bond type (kJ/mol) |
| H—C                                             | 3               | 413                               | .....                                          |
| C—N                                             | 1               | 305                               | 305                                            |
| N=C                                             | 1               | 615                               | .....                                          |
| C=O                                             | 1               | .....                             | 745                                            |

| $\text{H}_2\text{O}$ |                 |                                   |                                                |
|----------------------|-----------------|-----------------------------------|------------------------------------------------|
| Bond type            | Number of bonds | Energy to break one bond (kJ/mol) | Energy to break all of each bond type (kJ/mol) |
| H—O                  | 2               | 467                               | .....                                          |

total energy to break **all** bonds = ..... kJ/mol

- (ii) A similar calculation shows the energy released when bonds are formed in the products is greater than your answer above.

Complete the paragraph below by underlining the correct word(s) in the brackets. [3]

When a chemical reaction happens, energy is transferred to or from the surroundings. Energy is transferred to the surroundings if the energy released is **(less than / equal to / greater than)** the energy absorbed. This is called **(an endothermic / a combustion / an exothermic)** reaction. The temperature of the surroundings **(decreases / stays the same / increases)**.



- (iii) The Bhopal disaster was caused because of thermal runaway.

Explain what is meant by the term 'thermal runaway'.

[2]

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- (b) Spinach grown in north-eastern Japan was contaminated by radioactivity from the Fukushima nuclear power station accident in March 2011. Radioactive dust deposited iodine-131 atoms on spinach leaves. Its activity was 8200 units.

No spinach was allowed to be used for 96 days.

Iodine-131 has a half-life of 8 days.

- (i) The initial activity of the iodine-131 was 8200 units. Calculate the activity after 8 days.

[1]

activity ..... units

- (ii) Calculate the number of half-lives in 96 days.

[1]

number = .....

- (iii) Calculate the activity after 96 days.

[2]

activity ..... units

